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Statistical controversies in clinical research: scientific and ethical problems with adaptive randomization in comparative clinical trials

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Background: In recent years, various outcome adaptive randomization (AR) methods have been used to conduct comparative clinical trials. Rather than randomizing patients equally between treatments, outcome AR uses the accumulating data to unbalance the randomization probabilities in favor of the treatment arm that currently is superior empirically. This is motivated by the idea that, on average, more patients in the trial will be given the treatment that is truly superior, so AR is ethically more desirable than equal randomization. AR remains controversial, however, and some of its properties are not well understood by the clinical trials community.

Materials and methods: Computer simulation was used to evaluate properties of a 200-patient clinical trial conducted using one of four Bayesian AR methods and compare them to an equally randomized group sequential design.

Results: Outcome AR has several undesirable properties. These include a high probability of a sample size imbalance in the wrong direction, which might be surprising to nonstatisticians, wherein many more patients are assigned to the inferior treatment arm, the opposite of the intended effect. Compared with an equally randomized design, outcome AR produces less reliable final inferences, including a greatly overestimated actual treatment effect difference and smaller power to detect a treatment difference. This estimation bias becomes much larger if the prognosis of the accrued patients either improves or worsens systematically during the trial.

Conclusions: AR produces inferential problems that decrease potential benefit to future patients, and may decrease benefit to patients enrolled in the trial. These problems should be weighed against its putative ethical benefit. For randomized comparative trials to obtain confirmatory comparisons, designs with fixed randomization probabilities and group sequential decision rules appear to be preferable to AR, scientifically, and ethically.

Key words: adaptive randomization, Bayesian design, clinical trial, estimation bias, ethics, group sequential design

introduction

why randomize?

In medical research, the randomized comparative trial (RCT) is the scientific gold standard for obtaining confirmatory treatment comparisons. While the statistical rationale for randomizing is not well understood by many practitioners, it is accepted widely that randomizing patients between treatments is the right thing to do if one wants to obtain an answer to the question, 'Is one of these treatments better than the other?'

To see why randomization works when comparing two treatments, B and A, imagine one could make two copies of each patient, treat one copy with B, and treat the other copy with A. Suppose the primary outcome, response, is >50% shrinkage of a solid tumor within 12 weeks. If Y_B indicates response with B and Y_A indicates response with A, then $Y_B - Y_A$ is the *causal effect of B versus A* for that patient. The imaginary outcomes Y_B and Y_A are called *counterfactuals*. For example, if the copy of the patient that received B achieves a response and the copy that received A does not, then $Y_B - Y_A = 1 - 0 = 1$. The mean of all $Y_B - Y_A$ differences over a sample of such patient pairs would estimate the *causal effect of B versus A*. While one cannot make copies of patients, in practice a physician's choice between B and A for a given patient often involves thinking about this imaginary experiment.

To compare treatments for a patient population, the key parameters are $p_B = \text{Prob}(\text{response if a patient receives } B)$ and $p_A = \text{Prob}(\text{response if a patient receives } A)$, and $\Delta_{B,A} = p_B - p_A$ is the *population B-versus-A treatment*. For survival time, the parameters are mean survival times, m_A and m_B, and $\Delta_{B,A} = m_B - m_A$.

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While $\Delta_{B,A}$ is a conceptual object, its meaning is clear, since large positive values correspond to superiority of B over A, and conversely. This paper is about statistical methods for conducting a RCT to estimate $\Delta_{B,A}$ and decide whether one treatment is better than the other, and how randomizing adaptively rather than equally can affect such inferences.

If one randomizes patients equally between B and A, the difference between the sample means or proportions is an unbiased estimator of $\Delta_{B,A}$. The idea of unbiased estimation sometimes is misunderstood. Because any statistical estimator is a random quantity, an unbiased estimator does not *equal* $\Delta_{B,A}$, but rather follows a distribution having mean equal to $\Delta_{B,A}$. Figure 1 illustrates distributions of two estimators of $\Delta_{B,A}$, one obtained from a RCT and other from observational data. The RCT estimator is unbiased because its distribution is centered around $\Delta_{B,A}$, while the observational data estimator is biased because its distribution is shifted to the right, so it overestimates $\Delta_{B,A}$.

problems with observational data

Many things can go wrong with treatment comparison if one does not randomize. A common approach is to estimate $\Delta_{B,A}$ by collecting observational data on patients treated with either B or A, and compute the difference between the empirical response rates or sample means as the statistical estimator. This avoids the time and expense of conducting a clinical trial, but easily can lead to incorrect conclusions. If the patients who received B on average had better prognosis than those who received A, this can lead to *estimation bias*, denoted by $\text{bias}_{B,A}$. The usual difference between the empirical response rates does not estimate $\Delta_{B,A}$, but actually estimates $\Delta_{B,A} + \text{bias}_{B,A}$. For example, if the actual response rates are $p_B = 0.20$ and $p_A = 0.30$, the actual treatment effect is $\Delta_{B,A} = 0.20–0.30 = -0.10$, so A is slightly better than B.

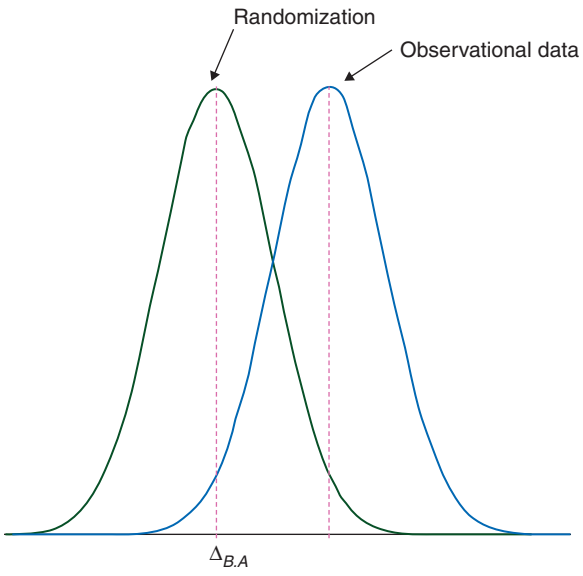


Figure 1. Distributions of estimators of the B-versus-A treatment effect $\Delta_{B,A}$ based on data from a randomized comparative trial, and an observational dataset that introduces bias by unequally favoring B over A due to nontreatment effects.

If $\text{bias}_{B,A} = 0.30$ due to better prognosis, on average, in the B patients, the difference between the estimated response probabilities will take on a value with mean $\Delta_{B,A} + \text{bias}_{B,A} = -0.10 + 0.30 = 0.20$ rather than $\Delta_{B,A} = -0.10$. This likely would lead one to believe, incorrectly, that there is a substantive advantage of B over A, be unrealistically optimistic about B, and organize a large phase III trial yielding negative results.

Sources of bias in observational data often are not obvious. One simply may not know that the patients who received treatment B tended to have better prognosis, and there often are unknown *latent variables*, that affect outcome and are not balanced between the two treatment groups. Latent variables may arise from systematic differences between nurses, physicians, or medical centers, changes in supportive care over time, different drop-out rates due to differences in toxicity between the treatments, or bias in selecting each patient's treatment, which may or may not be intentional. Such latent variables easily may cause a statistical estimator to be biased.

Many clinical trialists claim that bias in observational data can be corrected by fitting a statistical regression model, such as a logistic model for response or a Cox model for survival time, to adjust for effects of patient prognostic covariates. Unfortunately, *fitting a standard regression model will not correct for bias in observational data*. Valid statistical methods to correct for bias in treatment comparison based on observational data include inverse probability of treatment weighting [1], stratification, or case matching [2–8].

alternatives to equal randomization

While the unbiased comparison provided by randomization serves the needs of future patients, flipping a coin to choose each patient's treatment looks strange to many nonstatisticians. In medical practice, a physician chooses each patient's treatment based on the patient's prognostic characteristics. This is inherently biased, so a RCT may be considered the antithesis of routine medical practice. The physicians involved in a trial must have *equipoise* in that they have no reason to favor one treatment over the other. If the goal is to maximize benefit to the patients enrolled in the trial, then it may seem that the best action is simply to give each new patient the treatment that currently has the larger estimated response rate or mean survival time. This is called a 'greedy', 'play-the-winner (PTW)', or 'myopic' algorithm. Perhaps counterintuitively, a PTW algorithm may not be best for the patients in the trial. Consider a comparison of B and A in terms of response probabilities p_A and p_B, when the true (unknown) values are $p_A = 0.30$ and $p_B = 0.50$. Suppose one begins by treating three patients on each arm and thereafter uses the PTW rule but, by chance, 0/3 responses are observed with B, which has probability 0.125, and at least one response is observed with A, which has probability 0.657. Since the empirical response rate with B is 0 and the empirical response rate with A is positive, thereafter all future patients in the trial will be given treatment A. Thus, due to random variation, the trial gets stuck giving the inferior treatment to all but three patients. This phenomenon with greedy decision rules has been well known for many years in sequential analysis [9].

A compromise between equal randomization and PTW is adaptive randomization (AR), which uses the observed data

from previous patients in the trial to compute treatment randomization probabilities for newly accrued patients. These sometimes are called 'randomized play-the-winner' designs [10, 11]. For example, if one response were observed in three patients treated with A, and no responses were observed in three patients treated with B, an AR method might randomize the next patient to B with probability 0.25 and to A with probability 0.75, thus allowing additional data to be obtained on treatment B. The usual argument motivating using AR in place of equal randomization is that it is ethically more desirable because, on average, more patients are given the superior treatment. The goal of AR is to obtain sample sizes for the two arms, N_A and N_B, unbalanced in favor of the arm having larger true success probability, to obtain $N_B > N_A$ if B has the larger response rate or longer mean survival time. Many practitioners consider AR to be a panacea for the ethical dilemma posed by randomized trials. The use of AR in clinical trials remains controversial, however [12–15, 21, 27].

There are many ways to do AR [11, 16–19]. We will discuss simple Bayesian methods, for binary response outcomes and event time outcomes, two common cases arising in practice. For the binary case, our focus will be a trial with up to 200 patients to compare treatments B and A having response probabilities p_B and p_A. Our AR methods are similar to those studied previously [20, 21], with AR probabilities computed using the posterior probability that B has a higher response rate than A, denoted by $q_n(A < B)$. The original version of AR, introduced by Thompson [22], randomizes patients to B with probability $q_n(A < B)$ and to A with probability $1 - q_n(A < B)$. We refer to this method as AR (1). A modification, which we will call AR(1/2), adjusts these probabilities so both are closer to 0.50. For example, if $q_n(A < B) = 0.75$ and $1 - q_n(A < B) = 0.25$, AR(1/2) turns these values into randomization probabilities 0.63 and 0.37. As a comparator, we consider a Bayesian group sequential (GS) design that randomizes patients equally between B and A, with a treatment comparison rule applied at $n = 50$, 100, 150, and 200 patients, and patients randomized in blocks of size 8 to avoid interim sample size imbalance. The rule decides whether to stop and conclude B is superior to A, or A is superior to B, or continue the trial. We refer to this design as 'Equal GS'. Details are given in the Appendix section.

achieved sample size distributions, bias, and power

Any type of random treatment assignment produces random sample sizes, N_B and N_A, with distributions depending on the trial design and randomization method used. Properties of AR methods derived from computer simulations often are reported in terms of mean sample size distributions. The mean, or 'expected' difference between the achieved sample sizes, $E(N_B - N_A)$, often is used to quantify a putative advantage of an AR method over equal randomization, which has mean 0. If B actually is better than A ($p_B > p_A$), then AR should allocate more patients to B than to A. This could be quantified by $E(N_B - N_A)$ being positive, but reporting only mean sample sizes is misleading, because AR produces much more disperse sample size distributions compared with equal randomization.

For all simulations, each case was replicated 10 000 times. Table 1 gives simulation results for a trial run using each of AR (1), AR(1/2), and the Equal GS design. In the null case where $p_B = p_A = 0.25$, by symmetry the false-positive (type I error) probability for each method is twice the tabled value 'Probability conclude $p_B > p_A$', so the false-positive probabilities are 0.18 for AR(1), 0.24 for AR(1/2), and 0.048 for Equal GS. The simulations show that both AR(1) and AR(1/2) have much larger false-positive rates compared with the Equal GS design, and smaller power when B is superior to A, with $p_B = 0.45$. The means of $N_B - N_A$ are quite large for the AR methods when $p_B = 0.35$ or 0.45 and $p_A = 0.25$. Large mean sample size differences, such as those in Table 1, are the most commonly used rationale for arguing that AR is ethically superior to equal randomization.

The means do not tell the whole story. For example, when $p_B = 0.35$ and $p_A = 0.25$, *AR(1) has a 14% chance of producing a sample size imbalance of 20 patients or more in favor of the inferior treatment*, the opposite of the claimed ethical advantage of AR. This apparently anomalous behavior is explained graphically by Figure 2, showing the distributions of $N_B - N_A$ for AR(1) and AR(1/2) in this case. Because AR probabilities, which are computed from interim data, are highly variable, the distributions of $N_B - N_A$ for AR(1) and AR(1/2) have large left tails below 0, corresponding to values of N_B much smaller than N_A. *The practical and ethical point is that AR may behave pathologically in that it carries a nontrivial risk of creating a large sample size imbalance in favor of the inferior treatment.* Reporting only mean sample size differences 66 for AR(1) and 37 for AR(1/2) thus is very misleading. This example illustrates two important points. Since N_B and N_A are highly disperse with AR, the mean

Table 1. Comparison of designs for a binary response outcome with maximum trial sample size $N = 200$ and true $p_A = 0.25$

p_B	Method	Probability conclude $p_B > p_A$	Mean (2.5th, 97.5th) percentiles of $N_B - N_A$ = sample size imbalance in favor of B over A	π_{20}	Bias
0.25	AR(1)	0.09	0 (−186, 186)	0.43	0.00
	AR(1/2)	0.12	0 (−100, 100)	0.33	0.00
	AR(1)*	0.025	1 (−186, 186)	0.45	0.00
	AR(1/2)*	0.025	−1 (−132, 132)	0.40	0.00
	Equal GS	0.024	0 (0, 0)	0.00	0.00
0.35	AR(1)	0.30	66 (−164, 188)	0.14	0.05
	AR(1/2)	0.41	37 (−50, 116)	0.07	0.06
	AR(1)*	0.14	83 (−182, 152)	0.14	0.05
	AR(1/2)*	0.13	64 (−68, 152)	0.08	0.03
	Equal GS	0.34	0 (−2, 2)	0.00	0.02
0.45	AR(1)	0.59	80 (−62, 184)	0.05	0.09
	AR(1/2)	0.80	38 (−7, 116)	0.01	0.10
	AR(1)*	0.35	111 (−70, 192)	0.04	0.09
	AR(1/2)*	0.40	87 (2, 158)	0.01	0.07
	Equal GS	0.86	0 (−2, 2)	0.00	0.04

Equal GS, Equal randomization was done in blocks of size 8 for the group sequential design. π_{20} = probability of a sample imbalance of 20 patients or more in the wrong direction. Bias = (estimated $\Delta_{B,A}$) − $\Delta_{B,A}$.

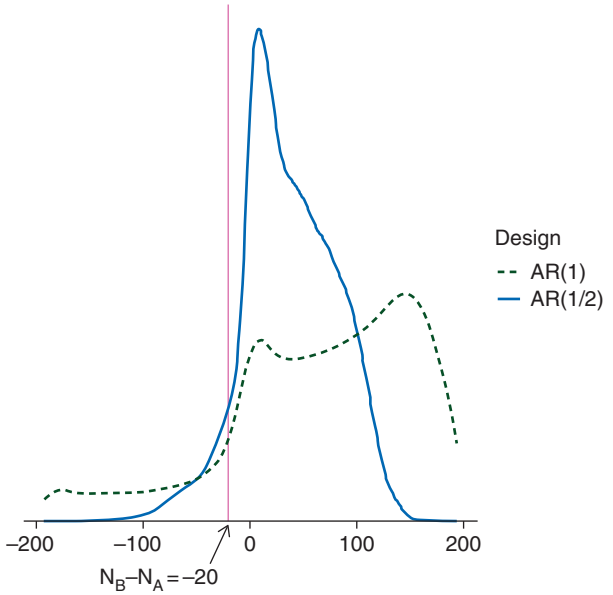


Figure 2. Distributions of the achieved sample size difference $N_B - N_A$ for a 200-patient trial conducted using either AR(1) or AR(1/2), when $p_A = 0.25$ and $p_B = 0.35$.

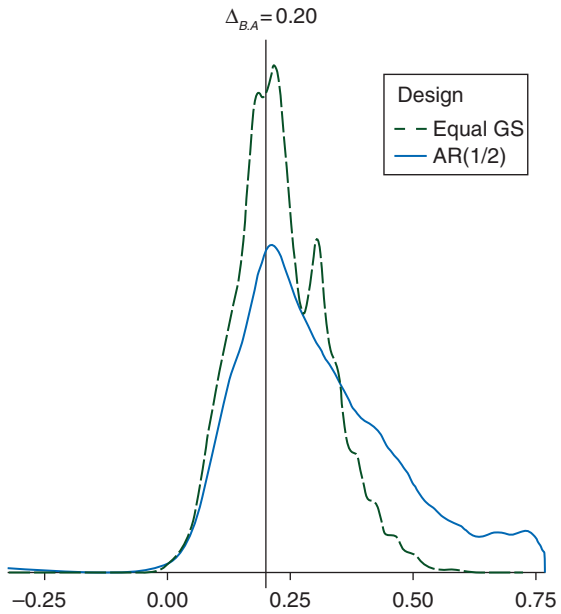


Figure 3. Distributions of the final estimator of the *B*-versus-*A* treatment effect $\Delta_{B,A} = 0.20$, for true response probabilities $p_A = 0.25$ and $p_B = 0.45$, from a 200-patient trial conducted using either a GS design with equal randomization or AR(1/2).

of $N_B - N_A$ does not adequately describe a given AR method's behavior, and the general claim that any AR method always is ethically superior to equal randomization is false.

While all adaptive designs introduce bias in final estimators, the bias is much larger when AR methods are used. This involves the ideas of accuracy and precision. Higher accuracy of an estimator is quantified by smaller bias, whereas higher precision means smaller variability. Figure 3 illustrates distributions of the estimators of $\Delta_{B,A}$ for the Equal GS and AR(1/2) designs, when $p_A = 0.25$ and $p_B = 0.45$, so the true effect is $\Delta_{B,A} = 0.20$. The distributions in Figure 3 do not have the symmetric, bell shapes as given in Figure 1, but rather are highly asymmetric, and shifted to the right, corresponding to overestimation of $\Delta_{B,A} = 0.20$. Both AR estimators are inaccurate. The estimator's distribution for AR (1/2) is more disperse, and less precise than that obtained from Equal GS. The mean estimate of $\Delta_{B,A}$ is 0.30 for AR(1/2), so its *Bias* is 0.10 (50%), the estimate is 0.23 for the Equal GS design, so its *Bias* is 0.03 (15%). *Using AR(1/2) triples the bias in the final estimator of the comparative treatment effect, compared to the Equal GS design.* Figure 3 also shows that the AR(1/2) estimator will be 0.40 or larger, more than double the true value $\Delta_{B,A} = 0.20$, with probability 0.25, compared with probability 0.05 with Equal GS. *Thus, at the end of a trial conducted using AR(1/2), in this case there is a 25% chance that one will overstate the actual benefit of treatment B over A by two fold or more.*

There is a substantial statistical literature on methods to correct for bias following a GS trial [23–26]. Unfortunately, these methods seldom are used in practice.

As explained in the Appendix, for these AR designs *B* is declared superior to *A* if the posterior probability $\text{Prob}(p_A < p_B | \text{data}_n) > 0.99$, and *A* is declared superior to *B* if this probability is <0.01. It may be argued that, to make the comparisons to Equal GS more fair, the cutoffs used in these stopping rules should be calibrated to obtain false-positive rates smaller than the values 0.18 for AR(1) and 0.24 for AR(1/2) in Table 1. If the cutoff 0.99 is replaced by 0.995 for AR(1), this ensures that the design will have type I error probability 0.05. We denote this design AR(1)*. Similarly, changing the cutoff from 0.99 to 0.9985 for AR(1/2) ensures type I error probability 0.05, and we call this AR(1/2)*. Table 1 shows that AR(1)* and AR(1/2)* both have very low power, however. When the true $p_B = 0.45$, the Equal GS design power = 0.86, AR(1)* has power = 0.35 and AR(1/2)* has power = 0.40. *Thus, if either AR decision rule is calibrated to have a type I error comparable with that of the Equal GS design, then the resulting AR design has much lower power to detect an actual treatment difference.* With these larger decision cutoffs, there are no changes in π_{20}, and bias still is substantial, so these problems persist.

Table 2. Mean and 95% intervals for total sample size, number of successes, and number of failures with the AR(1)*, AR(1/2)*, and the Equal GS designs, in the binary outcome case with true $p_A = 0.25$ and $p_B = 0.35$

Method	Mean (95% interval)		
	Total sample size ($N_A + N_B$)	Number of successes	Number of failures
AR(1)*	186 (33, 200)	60 (12, 80)	126 (22, 153)
AR(1/2)*	186 (44, 200)	59 (16, 77)	127 (28, 150)
Equal GS	179 (50, 200)	54 (15, 71)	125 (33, 152)

To quantify the actual advantage afforded by AR to the patients in the trial, Table 2 gives the means and 95% probability intervals (2.5th and 97.5th percentiles) of total sample size, number of successes, and number of failures for AR(1)*, AR(1/2)* and the Equal GS design, in the case where the true $p_A = 0.25$ and $p_B = 0.35$. On average, compared with Equal GS, these AR

methods give five or six more successes, and one or two more failures. The distributions of all three statistics are highly variable for all methods, however, so considering only mean values is very misleading. The small advantage in mean number of successes should be considered in light of the very small power figures for AR(1)* and AR(1/2)* shown in Table 1, which imply that the AR(1)* and AR(1/2)* designs are not likely to identify a true treatment advance and thus not likely to benefit future patients, and also their large estimation bias.

drift

A prominent argument against the use of AR is that it can lead to biased estimates in the presence of parameter drift. Drift occurs, for example, if p_A and p_B both increase by the same amount over time due to improving prognosis of patients enrolled over the course of the trial, but the comparative treatment effect $\Delta_{B,A}$ remains constant. Karrison et al. [17] discuss drift and recommend blocking to reduce potential bias due to drift. To examine the effects of drift, we re-simulated the cases in Table 1, but with p_A and p_B both increasing linearly from their initial values to final values 0.20 larger at the end of the trial. In each case, p_A was simulated from the true values $p_A(n) = 0.25 + 0.20(n/200)$ when n patients had been accrued. Thus, the treatment effect $\Delta_{B,A}$ remained constant throughout the trial, while both probabilities p_A and p_B increased over time. For example, in each case where nominally $p_B = 0.35$, outcomes for the B arm were simulated using the probabilities $p_B(n) = 0.35 + 0.20(n/200)$ when n patients had been accrued, so $p_B(0) = 0.35$ at the start of the trial and this drifted up to $p_B(200) = 0.55$ at the end the trial. Similarly, for arm A, $p_A(0) = 0.25$ at the start and $p_A(200) = 0.45$ at the end, but the difference was $p_B(n) - p_A(n) = 0.10$ throughout. The simulation results are summarized in the upper portion of Table 3. Drift has little effect on either the sample size distributions or π_{20}, but it increases estimation bias of both AR methods for true $p_B = 0.35$ or 0.45. In contrast, the bias for Equal GS is very small and does not change when drift is introduced. The largest effect of drift on the AR methods is that it causes them to conclude that $p_B > p_A$ with much higher probability in all cases. This increases the power when $p_B = 0.35$ or 0.45, but it also has the disastrous effect in the null case, where $\Delta_{B,A} = 0$, of increasing the false-positive rate of AR(1) from 0.18 to 0.36, and of AR(1/2) from 0.24 to 0.32.

With drift, the AR(1*) and AR(1/2)* designs have respective type I errors 0.20 and 0.10 compared with 0.08 for Equal GS, and both have power = 0.57 for $p_B = 0.45$, compared with power = 0.87 with Equal GS. All AR methods have bias that is at least triple that of Equal GS. *In the presence of drift, the AR(1)* and AR(1/2)* designs with type I error probability comparable that of the Equal GS design have actual type I error probability either about the same or more than double that of the Equal GS design, and greatly reduced power to detect the large improvement $p_B = 0.45$.* The values of π_{20} and magnitudes of bias are changed very slightly, so the problems with sample size imbalance in the wrong direction and large bias persist.

Since a total parameter drift of 0.20 over the course of a 200-patient trial may be considered large, we repeated the simulations with smaller total drift = 0.10, given in the lower portion of Table 3. With smaller overall drift, inflation of type I error for the AR methods is smaller, but the power figures are much smaller, specifically 0.45 for AR(1)* and 0.49 for AR(1/2)* compared with 0.86 for Equal GS at $p_A = 0.45$, and the problems with large π_{20} for true $p_A = 0.35$ and very large overestimation bias persist. A reviewer has noted that, in the presence of drift, theoretically, one can minimize the drift-induced inflation of type I error (and the drift-induced bias) in AR design by performing a block-stratified analysis. This would further reduce the power of the trial, however [13].

Table 3. Comparison of the designs in Table 1 when p_A and p_B drift upward together over time, corresponding to improving prognosis of enrolled patients

Initial value of p_B		Probability conclude $p_B > p_A$	Mean (2.5th, 97.5th) percentiles of $N_B - N_A$ = sample size imbalance in favor of B over A	π_{20}	Bias
Drift = 0.20					
0.25	AR(1)	0.18	0 (−174, 176)	0.43	0.00
	AR(1/2)	0.16	0 (−98, 96)	0.33	0.00
	AR(1)*	0.10	−2 (−188, 186)	0.45	0.00
	AR(1/2)*	0.05	0 (−128, 128)	0.40	0.00
	Equal GS	0.04	0 (0, 0)	0.00	0.00
0.35	AR(1)	0.47	56 (−142, 178)	0.14	0.08
	AR(1/2)	0.51	33 (−52, 110)	0.08	0.06
	AR(1)*	0.31	74 (−176, 192)	0.16	0.08
	AR(1/2)*	0.22	61 (−72, 148)	0.08	0.06
	Equal GS	0.36	0 (−2, 2)	0.00	0.02
0.45	AR(1)	0.74	66 (−72, 180)	0.06	0.11
	AR(1/2)	0.85	33 (−8, 110)	0.01	0.11
	AR(1)*	0.57	97 (−86, 194)	0.06	0.13
	AR(1/2)*	0.57	80 (1, 154)	0.01	0.09
	Equal GS	0.87	0 (−2, 2)	0.00	0.03
Drift = 0.10					
0.25	AR(1)	0.13	0 (−178, 178)	0.43	0.00
	AR(1/2)	0.14	0 (−98, 98)	0.33	0.00
	AR(1)*	0.05	−1 (−188, 188)	0.45	0.00
	AR(1/2)*	0.03	−1 (−130, 128)	0.40	0.00
	Equal GS	0.03	0 (0, 0)	0.00	0.00
0.35	AR(1)	0.38	61 (−154, 182)	0.15	0.08
	AR(1/2)	0.46	35 (−53, 114)	0.07	0.08
	AR(1)*	0.21	78 (−180, 192)	0.15	0.07
	AR(1/2)*	0.16	62 (−70, 148)	0.08	0.04
	Equal GS	0.35	0 (−2, 2)	0.00	0.02
0.45	AR(1)	0.67	0 (−69, 184)	0.05	0.13
	AR(1/2)	0.83	0 (−8, 114)	0.01	0.13
	AR(1)*	0.45	0 (−82, 194)	0.05	0.11
	AR(1/2)*	0.49	0 (2, 156)	0.01	0.08
	Equal GS	0.86	0 (−2, 2)	0.00	0.04

p_A and p_B drift from their initial values to final values either ($p_A + 0.20$, $p_B + 0.20$) or ($p_A + 0.10$, $p_B + 0.10$). All other parameters are as in Table 1.

event time outcomes

Because progression-free-survival time or overall survival (OS) time often are used as the primary outcome in RCTs, it is useful

to examine how similar AR methods applied with time-to-event outcomes behave compared with equal randomization. The following simulations focus on a trial comparing the median OS times of treatments B and A. We assume the commonly used Bayesian model in which the survival times in each arm follow exponential distributions, and their medians m_A and m_B follow identical inverse gamma priors, with means 12 months and variances 1000. This prior is equivalent to two observed events in ~13 months of follow-up, so it is minimally informative. In the simulations, all patients are followed until they have an event, and patient arrival times are simulated from a completely random (Poisson) process with an assumed accrual rate of 5 patients per month. For AR(1) and AR(1/2), it is concluded that $m_B > m_A$ if the posterior probability $\Pr(m_B > m_A|\text{data}) > 0.99$, and that $m_A > m_B$ if $\Pr(m_A > m_B|\text{data}) > 0.99$. To control type I error at 0.05, the cutoff 0.9968 was used for both AR(1)* and AR(1/2)*. For the Equal GS design, the randomization was restricted to achieve perfect balance between the two sample sizes when 50, 100, 150, and 200 patients had been enrolled, where the comparative tests were carried out. Additional details are given in the Appendix section.

Simulation results for the five designs comparing A to B in terms of OS times are summarized in Table 4. In null case where both median OS times are $m_A = m_B = 12$ months, both AR(1) and AR(1/2) have much larger false-positive rates of 0.14 compared with 0.04 for the Equal GS design. Unlike the binary case, for event time outcomes the AR methods produce slightly biased estimators in the null case, although this may be due to simulation variation. When B actually is superior to A, both AR methods produce large bias, roughly double that of the Equal GS design. For example, when $m_B = 16$ months, on average AR(1/2) gives estimated median survival 18.10 months, a 13% overestimate, compared with 16.78 months with the Equal GS design, a 5% overestimate. Thus, as in the binary outcome case, AR greatly overstates any actual advantage of B over A. When $m_B = 16$ months, the probability of a sample size imbalance of 20 patients or more in the wrong direction is $\pi_{20} = 0.11$ for AR(1) and $\pi_{20} = 0.07$ for AR(1/2). When B has a smaller advantage over A, with $m_B = 14$ months versus $m_A = 12$ months (not tabled), these figures are much larger, with $\pi_{20} = 0.22$ for AR(1) and $\pi_{20} = 0.16$ for AR(1/2). When type I error is controlled to ensure a more fair comparison, as in the binary outcome case AR(1)* and AR(1/2)* have greatly reduced power. For true medians $m_A = 12$ months and $m_B = 20$ months, AR(1)* has power = 0.46 and AR(1/2)* has power = 0.53, compared with power = 0.70 for Equal GS. The values of is π_{20} are unchanged, both AR* methods still have smaller bias, but they still overestimate the actual median survival for arm B by much more than Equal GS.

As a final simulation study, we evaluated the sensitivity of the five designs to accrual rate, for the case $m_A = 12$ months and $m_B = 16$ months. These simulations, summarized in Table 5, show that π_{20} increases with accrual rate for the AR methods. The AR(1)* and AR(1/2)* methods show a very similar pattern, but again with much smaller power. This is notable since more rapid accrual often is considered desirable in clinical trials.

Table 4. Comparison of the three designs with survival time outcomes, maximum $N = 200$, accrual rate 5 patients per month, and true median survival $m_A = 12$ months with treatment A

m_B (in months)		Probability conclude $m_B > m_A$	Mean (2.5th, 97.5th) percentiles of $N_B - N_A$ = sample size imbalance in favor of B over A	π_{20}	Bias (in months)
12	AR(1)	0.07	0 (−142, 142)	0.42	0.11
	AR(1/2)	07	−1 (−90, 90)	0.35	−0.06
	AR(1)*	0.024	0 (−156, 158)	0.43	0.02
	AR(1/2)*	0.025	−1 (−100, 100)	0.36	0.07
	Equal GS	0.025	0 (0, 0)	0.00	0.01
16	AR(1)	0.33	55 (−94, 154)	0.11	2.00
	AR(1/2)	0.36	36 (−48, 104)	0.07	2.10
	AR(1)*	0.17	69 (−102, 168)	0.11	1.09
	AR(1/2)*	0.21	46 (−54, 120)	0.08	1.22
	Equal GS	0.28	0 (0, 0)	0.00	0.78
20	AR(1)	0.66	64 (−26, 154)	0.03	4.00
	AR(1/2)	0.72	41 (−12, 102)	0.02	3.92
	AR(1)*	0.46	86 (−26, 172)	0.03	2.70
	AR(1/2)*	0.53	58 (−10, 124)	0.02	2.76
	Equal GS	0.70	0 (0, 0)	0.00	1.98

π_{20} = probability of a sample imbalance of 20 patients or more in the wrong direction. Bias = (estimated $\Delta_{B,A}$) − $\Delta_{B,A}$, in months.

Table 5. Evaluation of the effect of accrual rate on the three designs for trials with survival time outcomes, maximum $N = 200$, for the case where the true median survival times are $m_A = 12$ months and $m_B = 16$ months

Accrual rate (patients per month)		Probability conclude $m_B > m_A$	Mean (2.5th, 97.5th) percentiles of $N_B - N_A$ = sample size imbalance in favor of B over A	π_{20}	Bias (in months)
3	AR(1)	0.38	60 (93, 149)	0.09	2.14
	AR(1/2)	0.43	38 (−42, 106)	0.06	2.47
	AR(1)*	0.19	77 (−104, 174)	0.09	1.35
	AR(1/2)*	0.25	51 (−46, 126)	0.06	1.51
	Equal GS	0.35	0 (0, 0)	0.00	0.86
5	AR(1)	0.33	55 (−94, 154)	0.11	2.00
	AR(1/2)	0.36	36 (−48, 104)	0.07	2.10
	AR(1)*	0.17	69 (−102, 168)	0.11	1.09
	AR(1/2)*	0.21	46 (−54, 120)	0.08	1.22
	Equal GS	0.28	0 (0, 0)	0.00	0.78
10	AR(1)	0.25	48 (−102, 150)	0.14	1.86
	AR(1/2)	0.26	31 (−56, 100)	0.11	1.89
	AR(1)*	0.13	58 (−110, 162)	0.15	1.08
	AR(1/2)*	0.14	38 (−60, 116)	0.11	0.93
	Equal GS	0.20	0 (0, 0)	0.00	0.72

discussion

It appears that practitioners may not fully understand the properties of the AR methods that they have used. This may be due

to examining only average behavior, or not being aware of the large estimation bias and loss of power produced by AR. The use of AR methods in place of equal randomization in clinical trials remains controversial, and there has been an active debate in the medical literature in recent years [12–15, 20, 21, 27].

Our computer simulations have identified and quantified the following problems with AR with important ethical and scientific consequences.

(i) AR introduces much more variability into the distributions of the achieved total sample sizes N_A and N_B, compared with that introduced by equal randomization.
(ii) AR may have a high probability of a sample size imbalance in the wrong direction, which may be surprising to nonstatisticians.
(iii) AR introduces substantial bias in the estimator of the comparative treatment effect $\Delta_{B,A}$ by overestimating any actual effect. This bias becomes larger if there is parameter drift.
(iv) AR may produce a very large type I error rate. This problem becomes much more severe if there is parameter drift.
(v) If the decision cutoffs of AR are calibrated to ensure type I error 0.05, the resulting AR methods have much smaller power to detect true treatment differences compared with Equal GS.

The severity of each problem depends on the particular AR method used and design parameters. These problems with AR may be mitigated by adopting a fix-up strategy, such as stratifying or blocking, doing an initial 'burn-in' using equal randomization before applying AR, modifying the AR probabilities to shrink them toward 0.50, as AR(1/2), or correcting for bias in some *post hoc* manner. With such fix-ups, the resulting gain in the number of patients treated with the superior treatment over the inferior treatment, if in fact $\Delta_{B,A}$ does not equal 0, becomes much smaller.

We have not considered the multiarm case with more than two treatment groups. This case is much more complex, potentially involving both selection and testing, two or more stages, including a control arm or not, and many possible inferential goals. The issues discussed here should be examined when designing a multiarm randomized trial.

In conclusion, it is important to consider the actual gain with AR in light of the problems discussed here. Given our simulation results, and the simulation studies of others, scientific, and ethical problems introduced by AR must be weighed against its putative advantage.

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appendix

To define the adaptive randomization (AR) methods and statistical tests, denote the achieved sample size when an interim decision is made by n. For the case of binary outcomes, under a Bayesian model we assume that p_A and p_B follow independent beta (0.25, 0.75) priors. Bayesian AR probabilities were computed based on the posterior probability $q_n(A < B) = \text{Prob}(p_A < p_B|\text{data}_n)$. This is the posterior probability that, given the current data, B has a higher response rate than A. For fixed constant either $c = 1$ or $c = 1/2$, the AR method randomizes patients to B with probability

$$r_{n,B} = [q_n(A < B)]^c / \{[q_n(A < B)]^c + [1 - q_n(A < B)]^c\}$$

and to A with probability $r_{n,A} = 1 - r_{n,B}$. The version with $c = 1$, which we call 'AR(1)', was introduced by Thompson [22] and has AR probability $r_{n,B} = q_n(A < B)$. The version with $c = 1/2$, which we call 'AR(1/2)', is used commonly to shrink the AR probabilities toward 1/2, to avoid values of $r_{n,B}$ near 0 or 1. For both AR methods, the trial is stopped with B declared superior to A if $q_n(A < B) > 0.99$, or A declared superior to B if $q_n(B < A) = 1 - q_n(A < B) > 0.99$, and these rules are applied continuously. As a comparator, we used a group sequential (GS) Bayesian design that randomizes patients equally between B and A in blocks of size 8 to avoid interim sample size imbalance, with treatment comparison rule, applied at $n = 50$, 100, 150, and 200 patients, that stops and concludes that B is superior to A if

$$\Pr(p_A + 0.20 < p_B|\text{data}_n) > 0.95 - 0.80(n/200)$$

or that A is superior to B if the similar inequality holds with the roles of A and B reversed. If the trial is not stopped early, a final decision is made at $N = 200$. The decision cutoff parameters 0.95 and 0.80 were derived to ensure overall type I error probability <0.05. We refer to this design as 'Equal GS'.

For the case of time-to-event outcomes, the medians m_A and m_B were assumed to have inverse gamma priors with parameters (2.144, 13.728), which implies that m_A and m_B have prior mean 12 months and variance 1000. The criterion used to define $r_{n,B}$ and $r_{n,A} = 1 - r_{n,B}$ was $q_n(A < B) = \text{Prob}(m_A < m_B|\text{data}_n)$. The Equal GS trial was conducted using the decision rule to stop and declare B superior to A if

$$q_n(A < B) > 0.996 - 0.01(n/200).$$

with decision parameters 0.996 and 0.01 chosen to ensure overall type I error probability <0.05. Since, for each treatment arm $t = A$, B, the posterior of m_t is an inverse gamma distribution with parameters $(2.144 + \text{NE}_t,\ 13.728 + \text{TFU}_t)$, where NE_t = number of events and TFU_t = total follow-up time in arm t, the decision criterion at each analysis is computed using $q_n(A < B)$ based on the most recent values of $(\text{NE}_A, \text{TFU}_A)$ and $(\text{NE}_B, \text{TFU}_B)$.